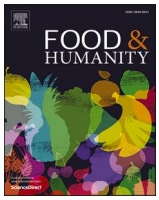
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Food waste in restaurants: A qualitative investigation of chefs' practices

Josephine Reardon [a], Kelly A. Way [b], M.E. Betsy Garrison [b,*]

[a] 5110 Herringbone Dr. Apt 233, Cincinnati, OH 45227, USA
[b] School of Human Environmental Sciences, 208 HOEC Bldg, University of Arkansas, Fayetteville, AR 72701, United States

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ABSTRACT

Through qualitative inquiry, the study's objective was to investigate the practices of chefs about food waste in restaurants. Informed by practice theory, 10 chefs in northwest Arkansas participated in semi-structured interviews. Five themes emerged from the data analysis: tracking, composting, repurposing ingredients, designing menus, and donating. The study highlights the importance of chef-driven initiatives in reducing food waste and concludes that it is imperative for restaurants to persist in endorsing food waste reduction methods. Communicating these types of initiatives to customers should not only increase their awareness, but also raise their environmental consciousness. Policymakers should enact simplified and lawful ways for the entire food service industry, including organizations interested in accepting food donations from restaurants, to reduce food waste, and thus, minimize their carbon footprints.

1. Introduction

The significance of reducing food waste lies in its environmental impact, as it prevents the disposal of food into landfills, mitigates methane emissions, and diminishes carbon footprints. Food waste is a global issue, both economically and environmentally, and a matter of public concern (Montesdeoca-Calderón et al., 2024; Stockli et al., 2018). Between 33 % and 50 % of all food produced globally is never eaten, and the value of this wasted food is worth over $1 trillion dollars (United Nations Environment Programme, 2024). The percentage of waste is strongly dependent on the kind of food. The FAO SAVE FOOD Initiative (2018) found that food waste comprised 20 % of beef, 35 % of fish products, and 45 % of fruits and vegetables. Western countries are the major contributors to the problem, with the U.S. producing the most food waste worldwide (Natural Resources Defense Council, 2022). In the U.S., about 40 % of all food is wasted, including 66 billion pounds of commercial leftover food. In the U.K., approximately 9.5 million tons of food waste are generated yearly, with 1.1 million tons (12 %) coming from the hospitality and food service sector. Australia wastes 7.6 million tons of food annually, costing their economy $36.6 billion. Boston Consulting Group estimates that global food waste represents $1.5 trillion in lost potential value (Mettler, 2023). These alarming statistics underscore the urgent need for collective efforts to reduce food waste in the food service sector.

Increasingly practice theory is being used to inform research about behavior change in hospitality, particularly related to environmental issues (Munir, 2022). Practice theory emphasizes the social and collective organization of practices and is considered grounded in real, often more tangible, processes (Hargreaves, 2011). Informed by practice theory, the study's aim was to address a gap in the research and contribute to the ongoing efforts in sustainable and responsible culinary strategies, fostering more environmentally conscious approaches within the restaurant industry.

2. Background

2.1. Causes of food waste

Bloom (2011) found that when both front-of-house and back-of-house waste is accounted for, around a half pound of food waste is created per meal served at a restaurant. ReFed (2023) found that full-service restaurants contributed 6.1 million tons of food waste that equaled 34 % of the total restaurant waste contribution at the remarkable cost of $67.6 billion dollars. Limited-service restaurants did not fare much better with an annual food waste of 2.6 million tons or 14 % waste contribution at a price tag of $25.3 billion dollars and lastly the "other foodservice" category disposed of 4.3 million tons (24 %) worth an estimated $44.7 billion dollars. Food waste in restaurants is not only indicative of the restaurant staff's actions, but also of the actions of consumers. Silvennoinen et al. (2015) found that the leading cause of

* Corresponding author.
E-mail address: megarris@uark.edu (M.E.B. Garrison).

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food waste in restaurants was from food that consumers did not eat.

According to the U.S. Bureau of Labor Statistics (2023), the average American consumer spends over $3000 annually on dining out, significantly contributing to the vast amount of food wasted in restaurants. Food Print (2024) estimates that 17 % of a diner's meal goes uneaten and that 55 % of restaurant leftovers are still edible. Furthermore, approximately 4 to 10 % of food purchased by restaurants is wasted before it even reaches the consumer. Key drivers of food waste in restaurants includes kitchen waste, serving loss, and plate leftovers (Heikkilä et al., 2016). This issue is exacerbated by portion sizes, which have increased significantly over the past 30 years, often being two to eight times larger than USDA or FDA standard servings (Food Print, 2024). Kitchen culture and staff behavior, such as improper food preparation and storage, failure to utilize food scraps and trimmings, overproduction of food, and unsafe practices also contribute to food loss (Principato et al., 2021).

2.2. Current strategies and initiatives

There is a growing body of literature focusing on food loss and waste, including zero waste cooking (Auguste Escoffier School of Culinary Arts, 2019), primarily examining various behavioral aspects (Aires et al., 2021; Bishop & Megicks, 2019; Nikolaus et al., 2018). Studies have explored consumers' attitudes toward and awareness of the high costs of food waste, based on rational choice theory or the theory of planned behavior (Aydin & Yildirim, 2021; Bishop & Megicks, 2019; Coşkun & Özbük , 2020; Graham-Rowe et al., 2015; Russell et al., 2017) and found that positive emotions were associated with environmental sustainability and can reinforce waste-reducing habits.

Previous research about the impact of purchasing large packages and overbuying (Brennan et al., 2023; Ganeson et al., 2023; Halloran et al., 2014;) and plate waste (Ergül et al., 2023; Gómez-Talal et al., 2024; Kamalul Ariffin et al., 2023; Levesque et al., 2023; Lorenz et al., 2017) on food waste has included strategies to reduce overbuying by introducing new menus and purchasing less food, improving storage methods, and reusing leftovers (Alcorn et al., 2021). Despite the significant amount of waste generated, there remains a lack of optimal waste management strategies,

Organizations have made efforts to mitigate food waste. In Berkeley, California, a study revealed that 65% of restaurants monitored food waste, with 84% of them utilizing compost bins to dispose of inedible food waste (Sakaguchi et al., 2018). Another study by Duursma et al. (2016), focusing on an individual restaurant, aimed to gain insight into the quantity and types of wasted food. They found that kitchen staff required better guidance and awareness regarding the importance of accurate measurement. Similarly, Sakaguchi et al. (2018) observed a lack of knowledge among restauranteurs regarding proper food waste measurement techniques.

ReFED (2018) developed a Restaurant Food Waste Action Guide, which outlines action-oriented solutions, tools, and best practices for effectively reducing food waste. Burgess (2016) conducted a study involving Michelin-starred chefs and discovered that their strategies for reducing food waste included eliminating à-la-carte menu items, offering smaller dish selections, and introducing three-course menus or tasting menus. Similarly, Gallion (2018) found that some local eateries increased the number of daily specials to allow chefs to repurpose surplus ingredients that would otherwise be discarded. Although the utilization of food scraps is not yet a widespread practice, there have been incremental efforts to incorporate this approach into restaurant operations, engaging kitchen staff in the process.

Controlling portion sizes is a chef practice that can effectively reduce food waste; however, the size of the plate also plays a crucial role. Wansink and van Ittersum (2013) observed that diners using larger plates served themselves 52% more and wasted 135% more food compared to those using smaller plates. Resnick and Belluz (2018) observed that consumers given larger bowls tended to take 16% more cereal than those provided with smaller bowls. Similarly, Ravandi and Jovanovic (2019) discovered that reducing plate sizes from large to small resulted in a reduction of plate waste by up to 30%. The trend towards larger portion sizes is identified as a primary driver of increased food waste costs for restaurant operators (Lipinski et al., 2013) and larger portion sizes have been found to contribute to the rising obesity rates in the U.S. (Herman et al., 2016).

The Food Waste Action Plan (2023) proposed a strategy of offering smaller portions of standard menu items alongside a variety of side dishes to better match guests' consumption preferences. Berkowitz et al. (2016) discovered that serving these smaller portions resulted in significantly lower levels of food waste, food costs, and waste disposal costs. Blum (2020) discerned that nearly 45% of restaurant leftovers were taken home in a to-go container or doggie bag and that social norms and embarrassment were driving factors in take home food decisions. Another factor was diners concern for the environment (Hamerman et al., 2018; Liao et al., 2018; Sebbane & Costa, 2018; Sharma et al., 2021).

2.3. Challenges, barriers and opportunities

Addressing food waste in restaurants requires a multifaceted approach involving both systemic changes and individual actions. Filimonau et al. (2020) determined that food waste programs in the UK and Netherlands are unviable and recommended a top-down approach with corporate-level involvement and routine top management-level monitoring. Researchers, such as Sakaguchi et al. (2018), have found that 75% of surveyed restaurants cited uncertainty about liabilities as a barrier to donating surplus food. Tax policies can make these donations costlier than discarding the food. The biggest liability for restaurants regarding donating food is the fear of being sued. Despite good intentions, many establishments hesitate to donate surplus food due to concerns over potential legal repercussions if someone were to become ill from consuming donated items. This issue is exacerbated by the lack of information and accessible guidelines concerning safety protocols for food donations. In the U.S., most states lack specific provisions addressing food safety for donated foods, relying instead on the guidelines outlined in the Food and Drug Administration (FDA) Food Code. However, this food code does not include language pertaining to food donations (Leib et al., 2018).

More than 15 years ago, Federal Bill Emerson Good Samaritan Food Donation Act was passed to encourage the donation of food to non-profit organizations (Feeding America, 2019). However, it was not as effective as intended because it failed to clearly define key terms such as "good faith" and "food quality" (Munger, 2018). Additionally, while the act provides strict protection for food donors who donate to non-profit organizations, it did not protect individuals who allow food donors onto their property to recover food, a deterrent. The U.S. FDA 2022 Food Code has helped to reduce barriers to food donations by clarifying, for the first time, that food donations from retail food establishments are acceptable as long as proper food safety practices are followed.

Despite a growing commitment to reducing food waste, restaurants are constrained by their customers and their menus. For example, a restaurant might have sufficient quantities of an item today, but face shortages tomorrow. Additionally, the cost of donating leftover food is a concern, as it involves paying employees, transportation expenses, and containers for food transport. Some charities are so large that individual restaurant donations would have minimal impact. For instance, the Los Angeles Regional Food Bank (Jimbo, 2020) collects and distributes tens of millions of pounds of food annually to those in need, but lacks a facility to prepare or serve food on-site, making the transport and storage of prepared foods challenging. Arrangements for the delivery of food must also be made – one cannot just pick a shelter and drop off food.

Although there is an abundance of research about food waste, there is less research that has focused on food waste in restaurants and the practices of chefs. The research that does exist is typically about fine

dining or luxury experiences (Batat, 2020; Filimonau et al., 2023) or is primarily published by Filimonau and colleagues (Filimonau et al., 2000, 2022, 2024; Filimonau, & Uddin, 2021) who have advocated for more qualitative investigations of food waste generated in commercial or out-of-home kitchens and how it can be reduced or prevented. Chefs are considered indispensable in the operation of restaurants (Wu et al., 2023) and to the adoption of food-related practices in restaurants (Filimonau et al., 2023). Implementing sustainable practices in the kitchen starts with careful menu planning to ensure the efficient use of ingredients, minimizing the potential for waste. Chefs also prioritize the use of seasonal and locally sourced products, which often have a longer shelf life and reduce the carbon footprint. They are also responsible for training kitchen staff on proper food storage, portion control, and creative ways to repurpose leftovers into new dishes. By fostering a culture of sustainability, chefs can significantly reduce food waste, benefiting both the environment and the bottom line of the restaurant.

In summary, the reviewed literature highlighted the growing interest in minimizing food waste in the food service section. Research has underscored the environmental and economic impacts of food waste, driving the need for better waste reduction strategies in commercial kitchens. The review also emphasized the importance of policy support and industry collaboration, to achieve substantial progress towards food waste reduction goals. The objective of the study, informed by practice theory, was to investigate the practices of chefs about food waste in restaurants, an understudied population in the food service sector.

3. Methods

The researchers chose to perform a qualitative study to gain an in-depth understanding of the complex and sometimes nuanced practices of chefs regarding food waste in different types of restaurants, as recommended by Sezerel and Filimonau (2023), Filimonau et al. (2024) and Batat (2020). This approach allows for the exploration of attitudes, motivations, and practices in a detailed and context-rich manner that quantitative methods might not capture (Guba & Lincoln, 1994). By conducting interviews, the researchers gathered rich, narrative data that revealed the underlying reasons behind behaviors related to food waste. This method is particularly valuable for identifying insights that can inform more effective strategies and interventions for reducing food waste in the restaurant industry.

Semi-structured interview guides were used to ensure consistency while allowing flexibility for participants to express their thoughts freely. This design enabled the collection of detailed, context-specific data, facilitating a deeper understanding of the factors influencing food waste practices in different culinary settings. Thematic analysis was then conducted to identify common patterns and unique insights (Guba & Lincoln, 1994) providing a more comprehensive picture of reducing food waste within the restaurant industry. The study was approved by the University of Arkansas' Institutional Review Board (IRB; human subjects) as protocol #1911234785.

3.1. Sampling and description of the sample

The researchers employed a purposive sampling technique to select participants who could provide rich, relevant insights into the topic because generalizability is not a part of qualitative inquiry. This study recruited self-identified chefs at any type of restaurant in northwest Arkansas (NWA) who volunteered to participate through the Facebook page NWA Chefs & Culinary Collaborative. Initial contact with chefs was established through instant messenger, and they were invited to take part in the study. All that volunteered and affirmed the IRB-approved informed consent protocol were interviewed, totaling 10 chefs.

Although the sample size of the study was small, the researchers deemed it appropriate because of its homogeneity and the narrow scope of the study. The participants were all chefs from a similar geographic region and culinary background, a smaller sample size was sufficient to capture the shared experiences relevant to the research objective. The scope of the study was relatively focused, concentrating specifically on the practices of chefs regarding food waste management. This narrower scope allowed for a more targeted inquiry, making a smaller sample size adequate for obtaining meaningful insights (Vasileiou et al., 2018).

Four of the chefs worked in fine dining establishments, and none were currently employed at a fast food, café , or buffet restaurant. Most of the participants had been employed at their current restaurant for over six years. Surprisingly, almost half of the chefs reported having no educational degrees in their chosen profession (Table 1).

3.2. Data collection

An interview protocol of open-ended questions was developed to measure food waste reduction practices, including benefits challenges and opportunities (Reardon, 2020). The interviews were conducted within a span of three weeks during the fall months. Each interview, ranging from 15 to 48 min, with an average duration of 29 min, was recorded to ensure accurate capturing of the insights shared by the participants. Following the conclusion of the interviews, chefs completed a questionnaire comprising demographic inquiries related to both their personal backgrounds and the restaurants where they currently held the position of chef.

3.3. Data analysis

The researchers manually transcribed the audio recordings verbatim from each interview, without the use of software, for precise documentation and a more thorough analysis. This approach allowed for a meticulous and detailed examination of the data, ensuring accuracy in the process. By personally transcribing the interviews, the researchers more closely engaged with the content, facilitating a deeper understanding of the participants' experiences. Additionally, manual

Table 1
Restaurant Type, Position Title, Length of Employment, Educational/Culinary Background, and Number of Environmentally Friendly Restaurants of Participants.

Variable	*n*	%
Restaurant Type		
Fine Dining	**4**	**40.0%**
Casual Dining	2	20.0%
Fast Casual	1	10.0%
Contract Foodservice	1	10.0%
Fine Casual	1	10.0%
Several Types	1	10.0%
Position Title		
Owner/Chef	3	30.0%
Executive Chef	**5**	**50.0%**
Corporate Executive Chef	1	10.0%
Sous Chef	1	10.0%
Length of Employment		
Less than 1 year	2	20.0%
1 −5 years	2	20.0%
7 −8.99 years	**3**	**30.0%**
Over 9 years	**3**	**30.0%**

Variable	*n*	%
Educational/Culinary Background		
Culinary school degree	2	20.0%
Culinary school degree + other degree	1	10.0%
2-year college degree	1	10.0%
4-year college degree	2	20.0%
No degree, self-taught	**4**	**40.0%**
Environmentally Friendly Restaurants		
0	**4**	**40.0%**
1	2	20.0%
2	2	20.0%
3	1	10.0%
5 +	1	10.0%

transcription provided the flexibility to capture nuances in speech patterns and non-verbal cues that might have been missed by automated transcription tools. The transcripts were then analyzed by thematic coding, in which the data were transformed from general codes into themes (Williams & Moser, 2019). The final thematic categories were elucidated through descriptive quotes, emphasizing both the commonalities and the distinctions observed across participants. To maintain confidentiality and to protect the identity and workplace details of the chefs, gender-neutral names were assigned.

4. Results

4.1. Tracking

Most of the participants revealed the absence of a formal system for monitoring food waste. Among them, two participants mentioned relying on a more intuitive approach, noticing when items seemed to be depleting rapidly. Both participants said that they needed to track these higher dollar items for economic reasons and that they put more focus on tracking meats than they did with produce. Additionally, a couple of the participants specified that their tracking efforts were concentrated solely on high-value items, such as meats. One of the participating chefs, Sam, elaborated on the existing process for tracking meat:

We break down beef tenderloin, we'll take the full weight of fat, and then we'll trim it, and take the weight of that, and then convert it into percentages at the end of the day. We'll take the weight of the trimmed one and then take the weight of the other and that's kind of how we track the amount of food waste. And we do that with stuff like short rib, filet, tenderloin, stuff like that.

Almost half of the participants acknowledged engaging in some form of food waste tracking. Jamie further explained that they employed a monthly food waste tracking system:

We keep track during the week, or during the month, sometimes on a daily basis. So, if something goes bad, or something is about to go bad, we'll know … we have to figure out what to do with it … and then every month we do a food inventory and we match that versus sales and find out what our cost percentage is. And then we can see how significant of a problem that waste played into whether we have a high cost or a low cost. A lot of the times, you can see if you have been wasting a lot per month.

Devin, Kelly, and Riley, as participants, were working with a company that facilitated the measurement of their food waste and provided them with the quantity diverted from landfills. The practice of measuring food waste was closely linked to the chefs' decision to engage in composting. Pat discussed the workings of the composting company in their restaurant and clarified why they refrained from tracking any food waste electronically:

They give us the weight—they weigh it—how much they are picking up. But in terms of percentages, like food we served and food that gets composted, we don't track that. We've looked at software to be able to do that, but the reality of what type of restaurant that we are, we aren't a chain concept that has one menu that never changes. We are constantly evolving, cooking from scratch; so, it would be a data entering nightmare to keep up with one more thing.

4.2. Composting

Among the participants, half were actively composting stated that implementing composting had not posed significant challenges, with the majority considering it the easiest practice to adopt. Of these, most collaborated with existing composting programs, which supplied bins and bags and managed the pickup of compost from the restaurants. Riley elaborated on the simplicity of composting:

They take care of everything. They pick it up, they put it in a new bag.… And we just about fill it every week, so they come pick it up like once a week and it's just really seamless. It's a great way to make sure that we are being responsible.

Alex, Jamie, and Casey were actively involved in the ongoing discussion with composting companies. Alex had not initiated composting yet due to the company's ongoing research on how to implement it effectively, particularly since seafood was a predominant component of the dishes served at the restaurant. This chef expressed the intention to begin the process promptly once a viable solution was identified. Jamie (quoted below) and Casey both conveyed their plans to transition within the upcoming months, emphasizing the necessity for everyone to be on board for the execution to be successful:

All of the ideas that they have seem cool, but we just need to execute them properly. It's not going to be of any good if we can't get everybody to buy into this with our staff members and having somebody responsible for making sure it's done. I will argue realistically and ask if we are really going to do this or are we just talking about it. If we are going to do this, then fine. I want to support that. But I also don't want to get all set up and then next week, it is already not being used.

None of the participants expressed opposition to composting; the two chefs who were not engaged in composting had distinct reasons for their choice. Taylor attributed their decision partly to the nature of the contract food service business, highlighting a lack of "incentive" to reduce food waste. However, Taylor had previously initiated composting in other restaurants where they worked and mentioned practicing composting in their personal life. In contrast, Lane straightforwardly mentioned not having sufficient waste to warrant composting.

All the chefs that implemented composting discussed the ease of it and how there were not many problems with the implementation. Some even regarded composting as the easiest practice to implement in their restaurant.

4.3. Repurposing ingredients

All, but one, of the participants emphasized the significance of repurposing ingredients as a crucial practice in their respective restaurants. Whether incorporating a single ingredient into multiple dishes or crafting daily specials, most chefs actively sought to minimize waste. Alex outlined their process:

How do we not waste things? And for us, it's figuring out how do we reuse things in the kitchen? Celery for example. A head of celery … a stalk of celery, we don't use the very end of the celery; we cut it off. If we throw it in the trash, I mean, that's wasteful; that's food waste. What we do is we take … we have a bin and we put those things in there, and then we use those things to boil seafood. We make stocks. So those are just little bitty things that I do at my little restaurant to go farther. We get bread every day and when we cut the sandwiches, the ends of the breads off, we throw those into the bin and make bread pudding out of that. If I didn't save all of those little pieces, then each time I make bread pudding, I would have to use a new piece of bread each time. We try not to have a lot of waste, but if we do, just put it in the bin.

Alex's restaurant also bought ingredients it could use in multiple ways:

I use a boneless, skinless, 10 oz. chicken breast, always fresh, never frozen. Because … I use it in my gumbo, I strip it up for my chicken strips, I use it for my jambalaya, and I use it for the center of a plate, and I use it for my Po'boy. So, I have five uses for that one product, and it goes a long way for me.

Sam expressed the effort to repurpose ingredients, but underscored the importance of thoughtful consideration and pride in the creation of each dish:

We aren't just going to take a bunch of scraps and make a risotto out of it. We have to be more thoughtful than that. Again, my job as a chef is to be very creative and low waste as possible.

4.4. Designing menus

Many of the participants acknowledged the consideration of food

waste, with the majority of them providing detailed insights into the repurposing of ingredients. Within this theme, sub-themes of economics and food cost emerged. Participants specifically highlighted the impact of food waste on food costs, emphasizing its negative consequences if not managed effectively. Casey articulated this viewpoint:

I try to, and I think every chef does. Just because it affects your food cost if you don't. So, you try to get the most out of every product you possibly can and use it in different ways. Um, a lot of us have a thing where we don't like to use the same thing twice or three or four times on a menu, so we try to use it or hide it in other things. So, like it will add flavor to this dish and a very mild component, whereas, on another dish, it will be very potent and forward. So, we try to blend it in that way.

Devin added:

Mainly what we are trying to do is make more money, so squeezing your penny out of everything is important … but it also just happens to be good for the environment and we can coincide with that, which is pretty cool.

Jamie also mentioned how it was more of an economical reason to minimize waste than an environmental or social motivation:

You have to … economically: you can't just throw away food. Because your profit comes from the last steaks, not the first steaks. You're paying rent, you're paying for the meat, and the last three of them on that is actually what you make for that day.

Others found joy in being creative while designing their menus with a goal of minimizing waste. Sam stated:

So, something that I do for that is, I try to have a list of ingredients that are in season and then I try not to stray away from those ingredients. So, basically, I try to use the same seven ingredients in all dishes in different ways. So, that's not really thinking about it in terms of cutting out waste, but that's what it's doing.… We aren't going to order butternut squash for one dish and if we don't sell it, it's going to be thrown away. You order butternut squash for four different dishes that's utilized in four different ways. So, it kind of helps with creativity and it helps with food waste because it's not just sitting there; it's not just being used if that dish is not that popular…. I don't necessarily think, how can I not waste things, I think of it more as, how can I use everything. Which I guess is the same thing.

Many chefs implemented measures to minimize food waste during the cooking process. However, a significant challenge arose from the unpredictability of guest numbers during each restaurant shift, resulting in some edible food that could no longer be sold. This introduced a dilemma: chefs had the option to discard the surplus food, compost it, or choose to donate it.

4.5. Donating

Half of the participants confirmed their regular practice of donating leftover food whenever feasible, and among them, three chefs expressed motivation driven by the observation of less fortunate individuals in their vicinity. Alex stated:

When I drive from my house to the restaurant, I drive past the homeless shelter and every day, every morning, I see 50–60 people standing with their backpacks and their sleeping bags…. It really does hurt my soul that they went to bed without food. And as a restaurant chef, I love to feed people. So, I know there are people every morning in a three-mile area, who have a food insecurity. Food insecurity is right here.….

When inquiring about why participants opted not to donate, a couple of the participants explained that, given the newer status of their restaurants, their primary focus was on business development and building a clientele. Nonetheless, both chefs emphasized their willingness to consider donation practices once their businesses were better established.

Only one chef mentioned how liability was the reason they chose not to donate:

I think the health department would have a problem with us, you know reheating and reheating. I would be concerned with like … once something leaves our space, the variables to control it is very challenging and I don't want somebody to get sick and there to be a liability issue. So, it is unfortunate that, you know, that's the case, but that being said, I think the food being donated is being donated for an event or those types of things as opposed to going to local homeless shelters.

When questioned about their awareness of the Bill Emerson Good Samaritan Act of 1996, a chef revealed a lack of specific knowledge about it. On the other hand, Jamie, who was actively involved in donating, expressed the opinion that an act like this would not alter the potential risks associated with foodborne illnesses:

Regardless of the act, I would have a personal problem with delivering something to somebody that I didn't feel [was] safe…. So, there is … whatever act they provide … that they pass or whatever, doesn't change the possibility of a food-borne illness. If for instance, the food was held longer than the allotted time or wasn't held at the proper temp or wasn't handled properly…. You know, as a chef, you should know if it is able to be served or not.

5. Discussion

The objective of this study was to investigate chefs' practices about food waste in restaurants. The study addressed gaps in the current literature by providing detailed, qualitative data on food waste practices from those directly involved in food preparation (chefs). Based on a thematic analysis, the results provide insights into practical applications and real-world strategies, cultural and behavioral shifts, policies, and sustainability. The chefs' firsthand experiences and innovative practices offer a tangible connection to theoretical models. Their accounts of overcoming operational challenges and shifting kitchen cultures provide depth to understanding the behavioral and systemic changes required for successful food waste reduction (Aydin & Hamilton, 2021; Batat, 2020; Filimonau et al., 2020) and complement the existing literature, including previous research of consumers (Bishop & Megicks, 2019; Coşkun, A., & Özbük, 2020; Hamerman et al., 2018; Liao et al., 2018; Stöckli et al., 2018), together, cultivating a more comprehensive understanding about food waste, specifically, and sustainability, more generally, in the food service industry.

Food waste presents a multifaceted challenge with significant environmental, economic, and social implications. These challenges, based on the findings of the current study indicate the importance of tracking inventory and food waste, composting, repurposing ingredients, designing efficient menus, and donating leftover food. Effective inventory management is fundamental to reducing food waste (Principato et al., 2021; Read & Muth, 2021). Accurate tracking systems enable kitchens to monitor stock levels, expiration dates, and usage patterns, ensuring that food is used before it spoils. Implementing digital inventory systems can provide real-time data, facilitating better decision-making and reducing the likelihood of overbuying. Regular audits of food waste can help identify key areas where waste is most prevalent, allowing for targeted interventions.

Composting is a sustainable method to manage unavoidable food waste (Audette, 2018). By converting organic waste into nutrient-rich compost, kitchens can reduce the amount of waste sent to landfills, thus lowering greenhouse gas emissions. Composting not only supports waste reduction but also enhances soil health, promoting sustainable agriculture. Encouraging businesses to adopt composting practices can significantly contribute to a balanced economy.

Repurposing ingredients across multiple recipes is a creative and efficient way to minimize waste. By designing menus that incorporate versatile ingredients, kitchen staff can ensure that surplus food is utilized effectively (Alcorn et al., 2020). For example, vegetable trimmings, including stems, can be used to make stocks, and stale bread can be transformed into croutons or breadcrumbs. This approach not only reduces waste, but also enhances culinary innovation and resourcefulness.

Menu design plays a crucial role in food waste reduction (Aires et al.,

2012; Alcorn et al., 2020). Menus should be planned with waste minimization in mind, prioritizing seasonal and locally sourced ingredients that are less likely to spoil quickly. Portion control is another critical aspect; by offering customizable portion sizes, kitchens can cater to varying appetites and reduce plate waste. Using smaller plates or bowls is an effective strategy (Ravandi & Jovanovic, 2019; Resnick & Belluz, 2018). Similarly, for portion control and food waste reduction because it leverages psychological and visual cues that influence eating behavior (Berkowitz et al., 2016; Herman et al., 2016: Wansink & van Ittersum, 2013). Additionally, integrating zero waste cooking that utilizes all parts of ingredients can further decrease food waste.

Both individually and collectively, tracking, repurposing ingredients, and designing menus all can translate into significant cost savings. By minimizing overbuying and optimizing ingredient use food service businesses can lower their procurement costs.

Donating leftover food to charities and food banks can be an effective way to address food insecurity while reducing waste. Establishing partnerships with local organizations can ensure that surplus food reaches those in need instead of being discarded (Sakaguchi et al., 2018). Implementing proper food safety protocols is essential to maintain the quality and safety of donated food (Principato et al., 2021). By fostering a culture of giving, the food industry can play a significant role in supporting community well-being while enhancing a business' reputation for sustainability. Such acts may be appealing to eco-conscious consumers; thereby, potentially increasing customer loyalty and market share.

Informed by practice theory, the results highlighted tangible actions that chefs are taking to reduce waste. This practical knowledge is essential for translating theoretical concepts into effective, everyday practices, such as those mapped by Principato et al. (2021). Understanding these discrepancies can drive more effective policymaking and support tailored interventions that resonate with the everyday realities of food service professionals.

These insights are important as solutions are often proposed at a policy or industry level, focusing on systemic changes rather than specific operational adjustments. The participating chefs highlighted specific challenges they faced in their kitchens, such as managing portion sizes, dealing with unpredictable customer counts and menu selections, and educating staff about waste reduction. They also provided concrete solutions, including real-time inventory tracking, and partnerships for composting. By detailing specific, in-the kitchen challenges and solutions, the chefs' insights provide a more grounded understanding of what it takes to implement zero waste food practices in real settings.

5.1. Limitations and future research

One of the limitations of the current study is that results of qualitative investigations are not generalizable. Another limitation is that this particular study involved a single location in a single U.S. state. Future research should be conducted using qualitative, quantitative or a combination of both research methods. For example, the addition of kitchen observations as a mixed-method design would allow for the triangulation of results. Future research should also broaden the pool of interviewed chefs, further diversify the range of restaurant types, and enlarge from a single site by including different, countries, cities and states, other than NWA, a recognized progressive area in the culinary world. Last, future research should include restaurant staff, both front and back of the house, in addition to chefs as well as all aspects of restaurant operations, including, if separate, baristas, beverage and bar personnel.

5.2. Implications

Reducing food waste through strategies like tracking, composting, repurposing ingredients, designing menus, and donating leftover food has profound implications for both the foodservice industry and the general public. The foodservice industry could realize economic benefits in the areas of cost savings, enhanced efficiency, and sustainability credentials and recognition (Batat, 2020; Lévesque et al., 2023). Foodservice operations can experience operational benefits such as meeting regulatory demands for sustainability and waste management. Additionally, training staff in sustainable practices may foster a culture of stewardship and innovation, which can potentially enhance job satisfaction and improve employee retention.

Zero waste kitchens can enhance the dining experience for customers, showing how the general public directly benefits from foodservice operations that follow food waste guidelines. The general public is concerned about foodservice operations for several reasons including the environment impact (waste reduction and sustainability practices), social responsibility (community support and responsibility), health and safety (food and nutrition), economic considerations (cost efficiency and supporting the local economy), and lastly consumer awareness and education (transparency and mindful consumption of food).

Lastly, these findings can be used as educational material for culinary schools, hospitality management programs, and professional development courses. Aspiring chefs and food service managers can learn from the experiences of professionals, fostering a new generation of chefs and food service professionals committed to zero waste and sustainability.

6. Conclusion

The implementation of comprehensive food waste reduction strategies benefits the foodservice industry by lowering costs, enhancing operational efficiency, enhances profitability, and improves sustainability credentials. For the general public, these practices promote environmental stewardship, conserves resources, reduces landfill use, supports community food security, and encourages more mindful and sustainable consumption patterns. Together, these efforts contribute to a more sustainable food system, balancing economic, environmental, and social goals while creating a more sustainable and resilient food system.

CRediT authorship contribution statement

Josephine Reardon: Writing – original draft, Methodology, Data curation, Conceptualization. **Kelly A. Way:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **M. E. Betsy Garrison:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors report there are no competing interests to declare.

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